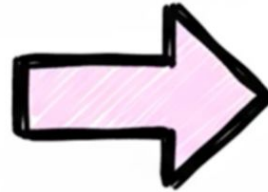


Numbers to Narrative

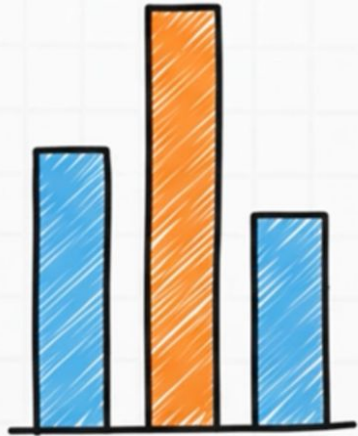


**Interactive
Earthquake Data
Visualization**



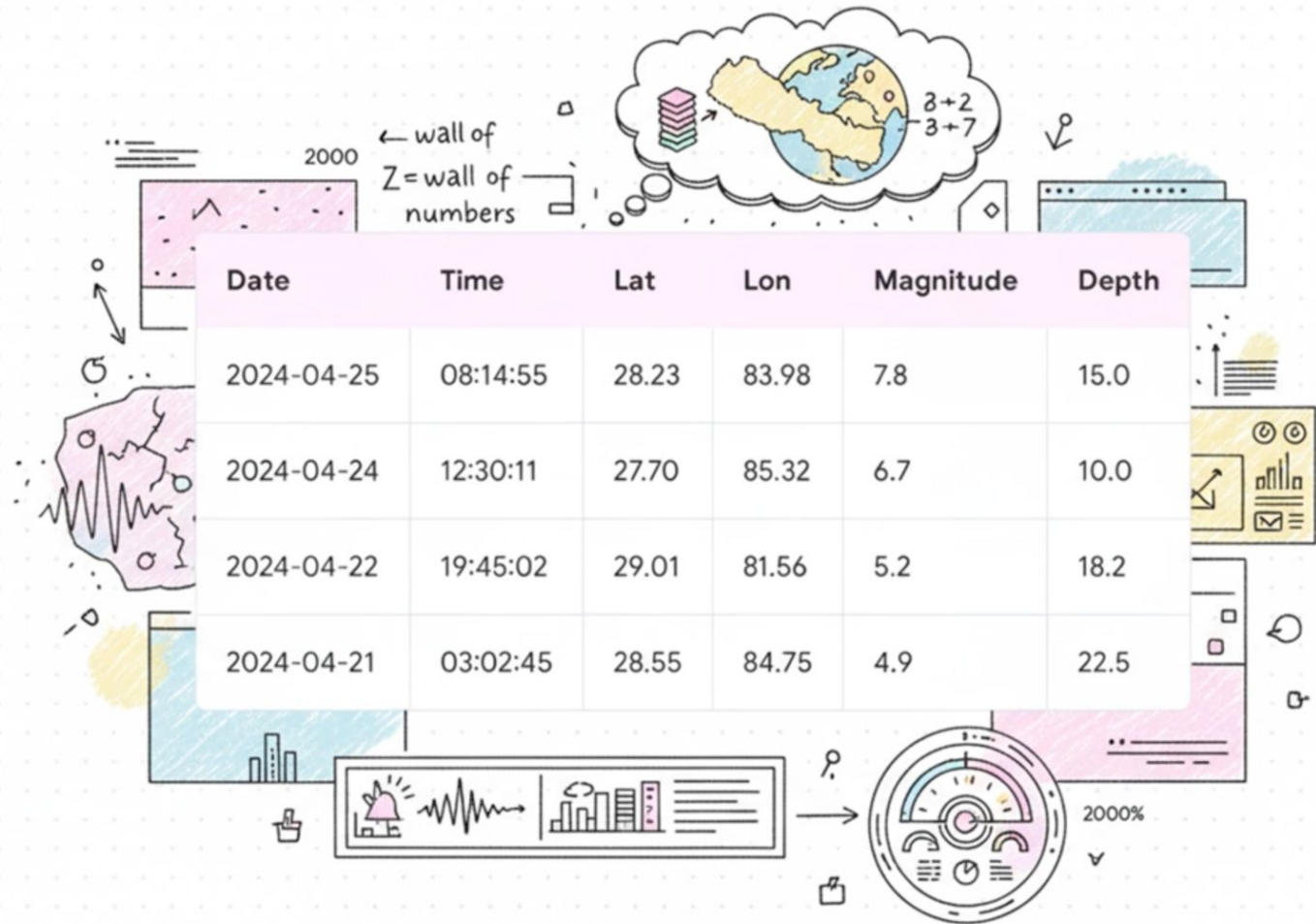
Python Bokeh

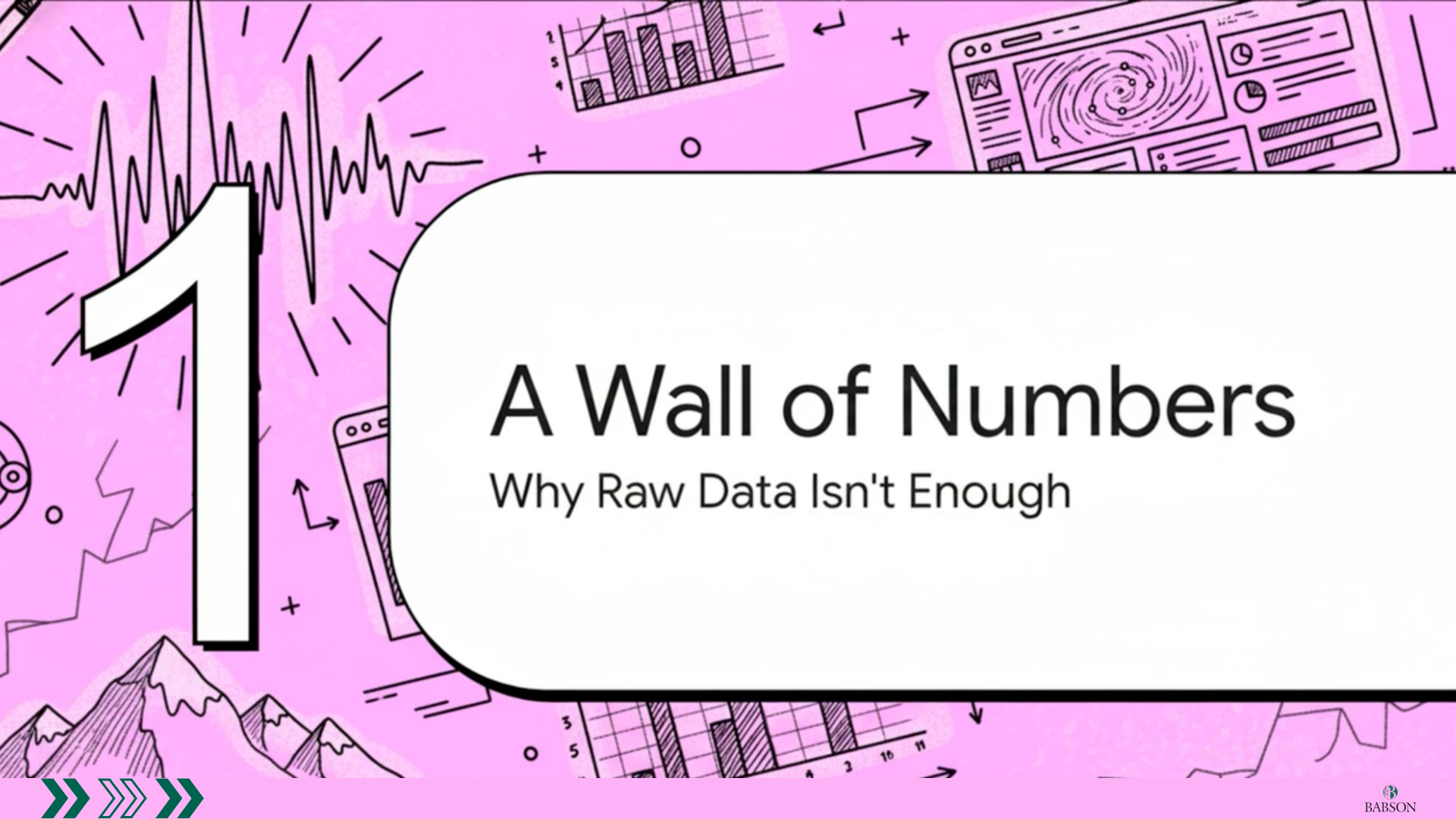
Presented by :
Jay Kishan Panjiyar



How do you find a **life-**
saving pattern in a
decade of data?





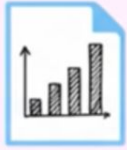


1

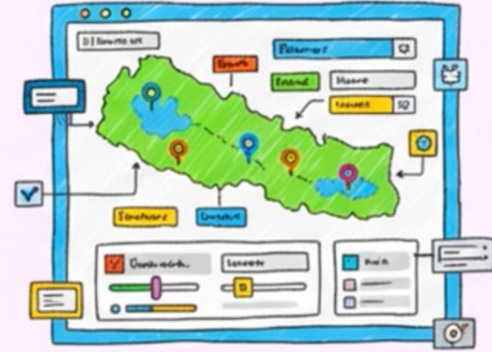
A Wall of Numbers

Why Raw Data Isn't Enough





Static Analysis: Raw CSV files that miss **dynamic** patterns and are **hard** to communicate.



Interactive Exploration: Dashboards revealing **time** & **location** trends for anyone to understand.

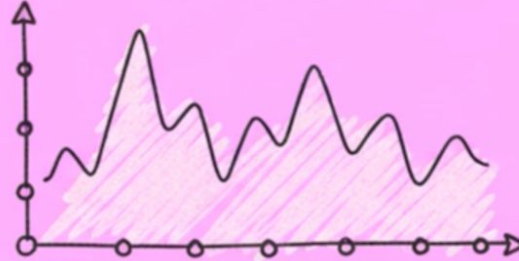


2

Coder's Storytelling Tool

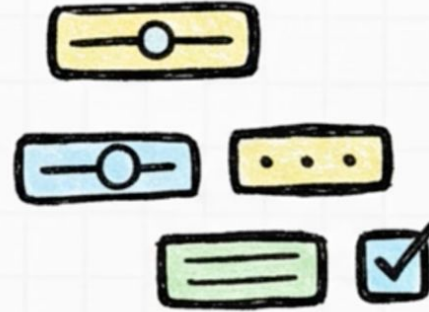
Introducing Bokeh 





Bokeh

A Python library for creating interactive visualizations and web apps, without needing JavaScript.



Why Bokeh?





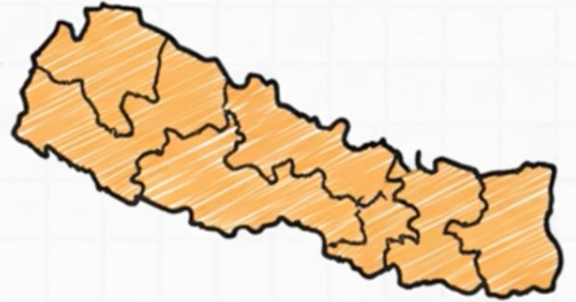
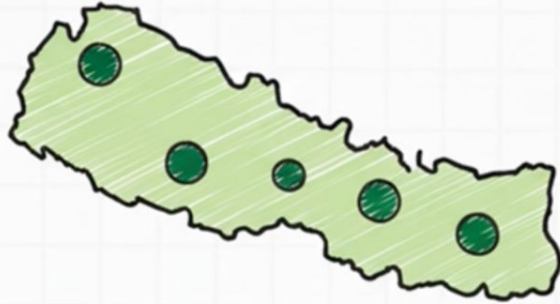
3

Building the Explorer

Ingredients and Recipe



The Three Core Datasets



Data Flow

1. Ingest

Raw CSV & GeoJSON files are loaded.



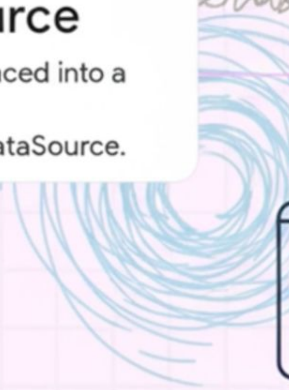
2. Process

The Pandas library cleans and organizes the data.



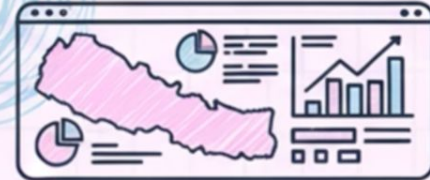
3. Source

Data is placed into a Bokeh ColumnDataSource.



4. Visualize

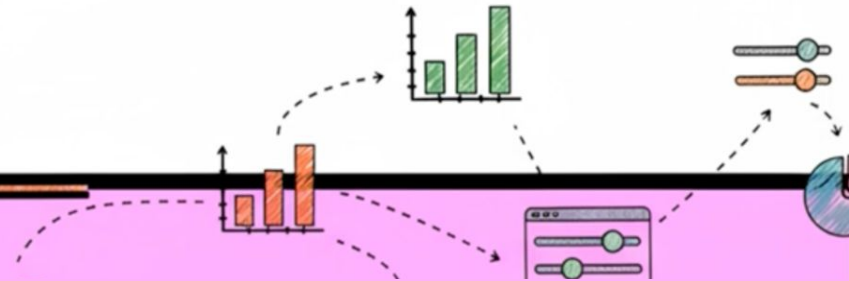
The map and charts are plotted from this central source.

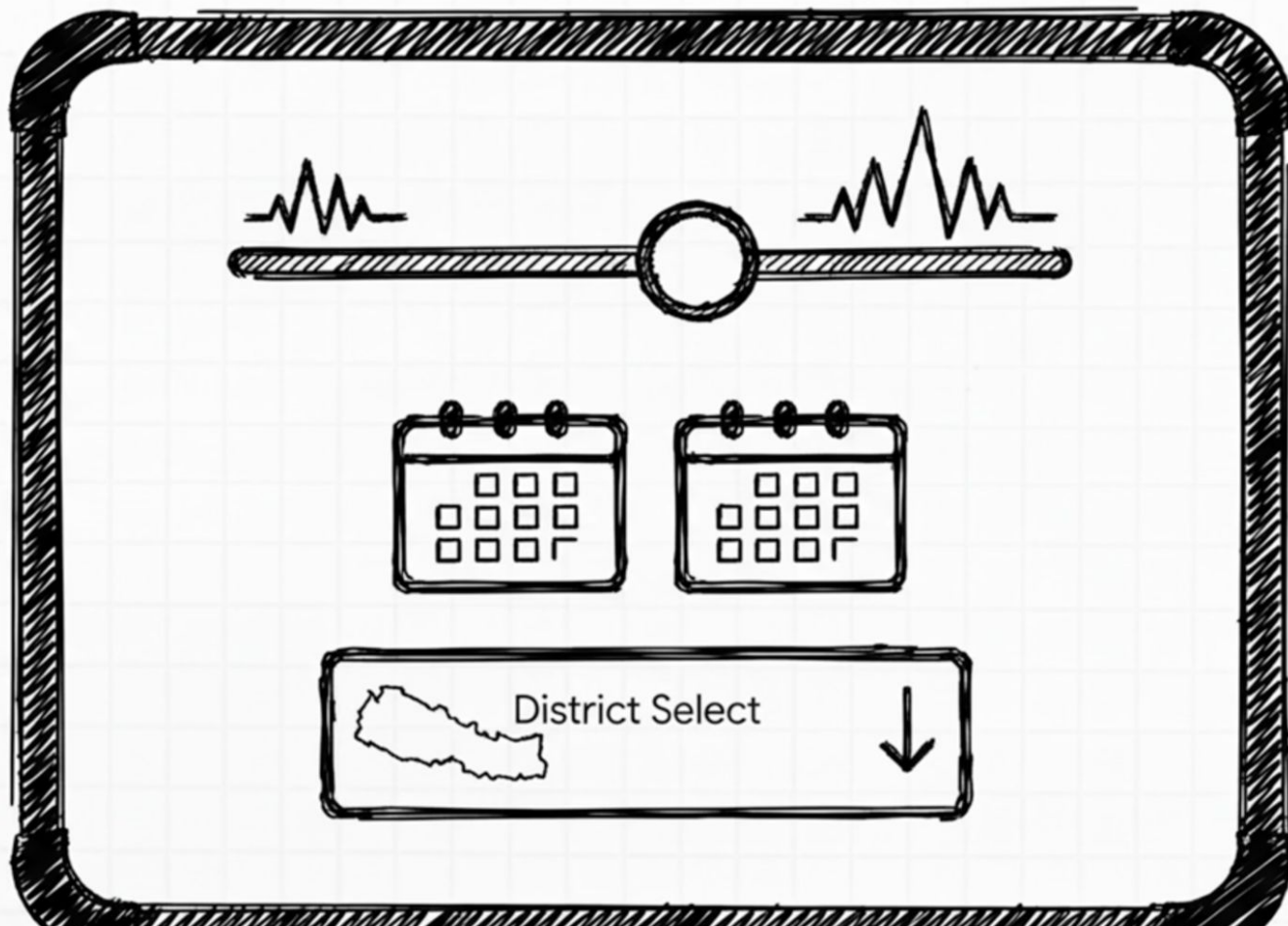


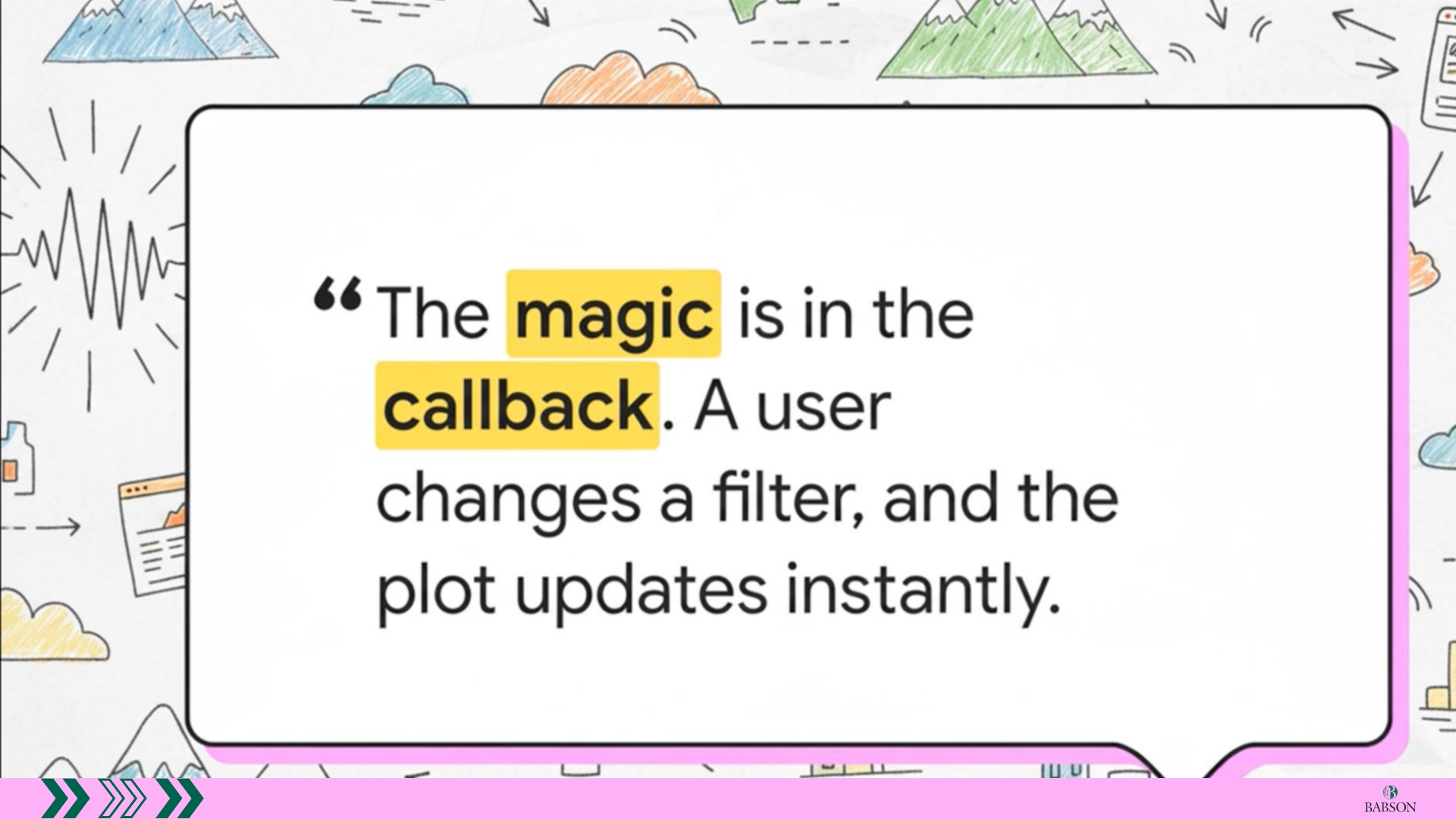
4

Data in Real-Time

Adding Interactivity







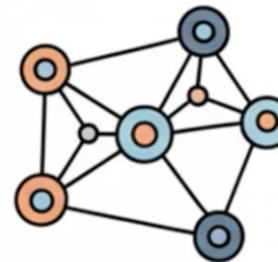
“The **magic** is in the **callback**. A user changes a filter, and the plot updates instantly.

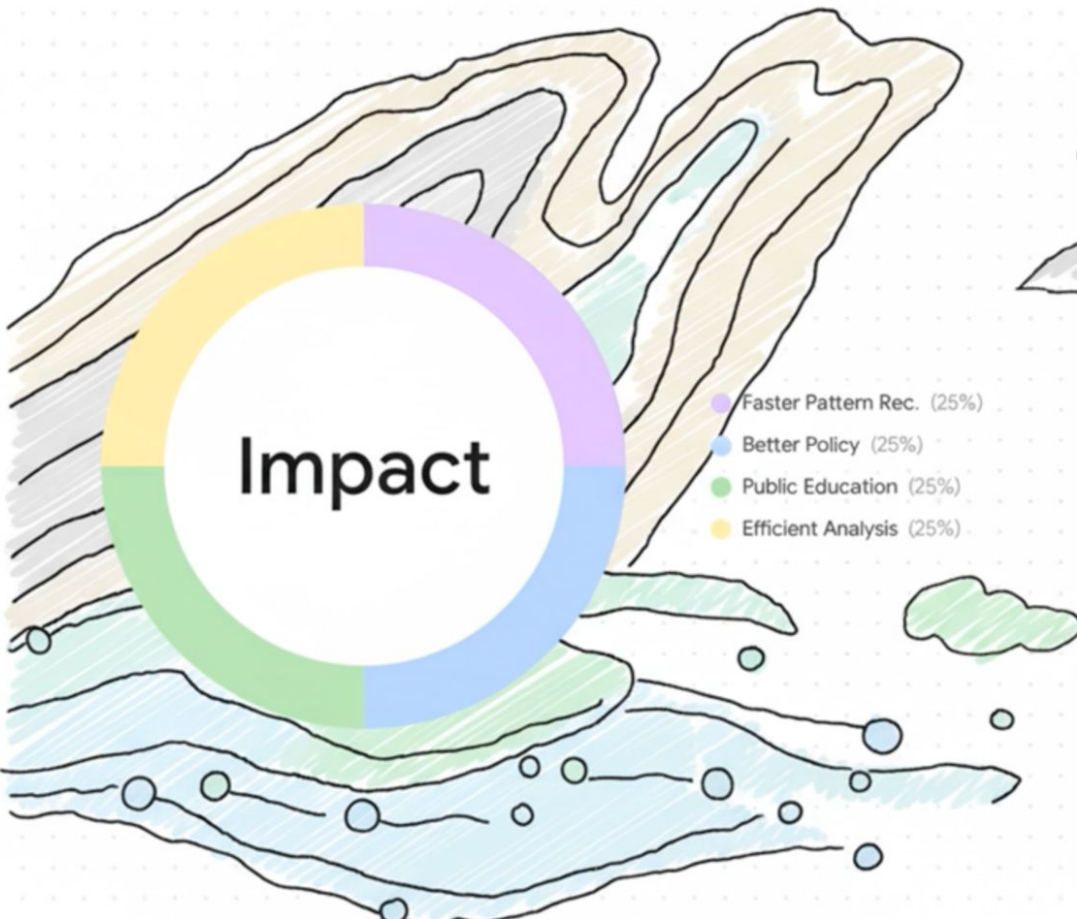


5

From Insight to Impact

Why This Matters





Impact

● Faster Pattern Rec. (25%)

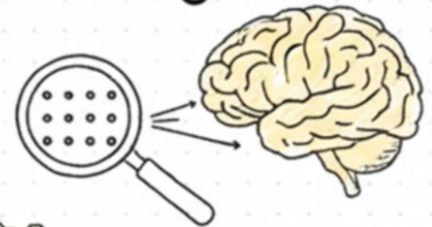
● Better Policy (25%)

● Public Education (25%)

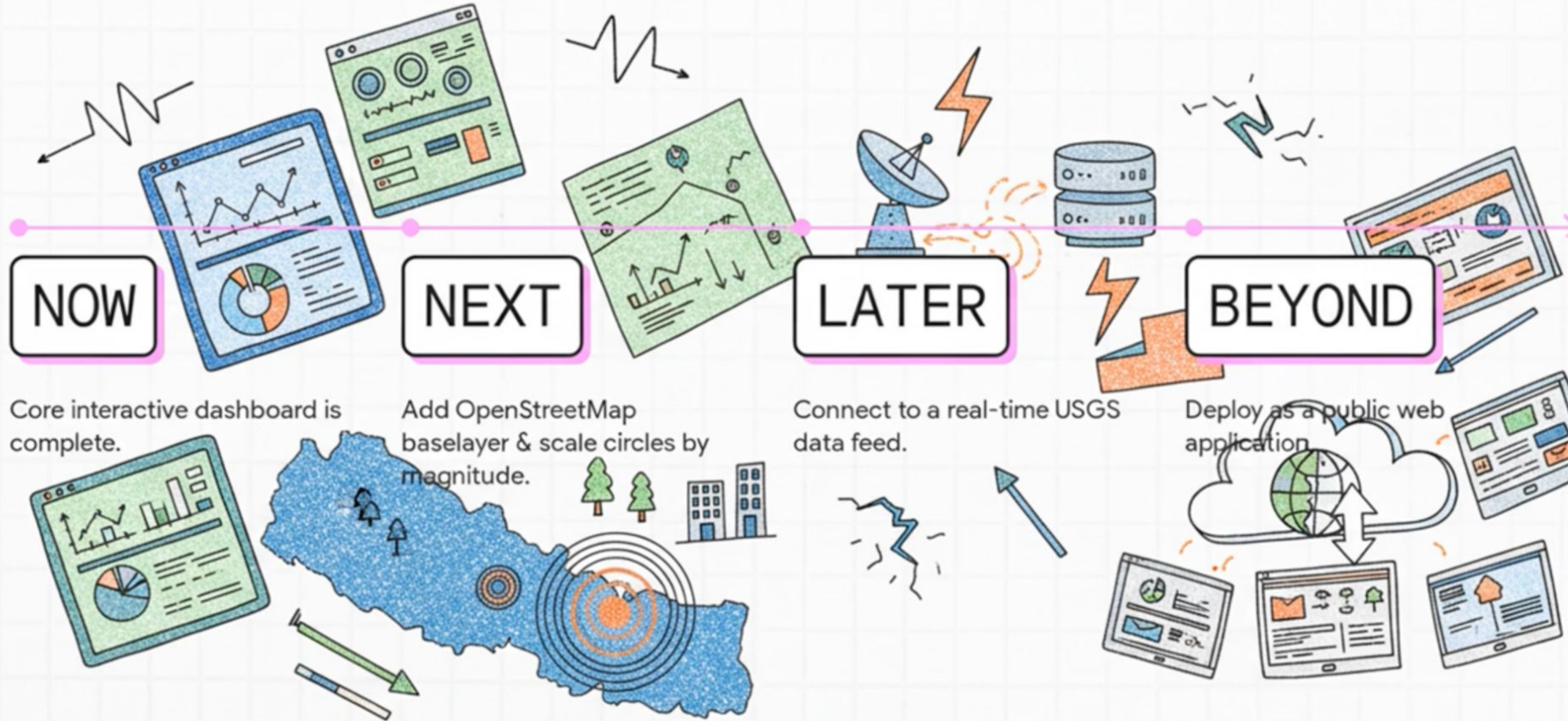
● Efficient Analysis (25%)



Empowering **scientists**
and **policymakers** with
clear, visual insights.



Future Enhancements





What **critical story** is
hidden in our **data**?





KERRY MURPHY HEALEY

BABSON
BEAVERS